Matrices and Systems of Linear Equations

In mathematics, the theory of linear systems is the basis and a fundamental part of linear algebra, probably the most important problem in mathematics is that of solving a system of linear equations. Well over 75 percent of all mathematical problems encountered in scientific or industrial applications involve solving a linear system at some stage. Computational algorithms for finding the solutions are an important part of numerical linear algebra, and play a prominent role in engineering, physics, chemistry, computer science, business and economics.

A ***system of linear equations*** is a collection of m equations in the variable quantities; $x_1, x_2, \dots \dots \dots, x_n$ of the form,

$$\begin{array}{c} a_{11}x_1 + a_{12}x_2 + \dots \dots \dots \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots \dots \dots \dots + a_{2n}x_n = b_2 \\ \dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots \\ \dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots \\ a_{m1}x_1 + a_{m2}x_2 + \dots \dots \dots + a_{mn}x_n = b_m \end{array} \qquad (1)$$

Where, the values of $\boldsymbol{a_{ij}}$ $(coefficients\ of\ variables)$, $\boldsymbol{b_i}(constants)$ and $\boldsymbol{x_j}$, $1 \le i \le m$, $1 \le j \le n$, are from the set of all real numbers.

The system (1) can be written in a matrix form; $\boldsymbol{AX = B}$, where,

$$\begin{pmatrix} a_{11} & a_{12} \dots \dots \dots \dots \dots & a_{1n} \\ a_{21} & a_{22} \dots \dots \dots \dots \dots & a_{2n} \\ \dots & \dots\dots\dots\dots\dots & \dots \\ \dots & \dots\dots\dots\dots\dots & \dots \\ \dots & \dots\dots\dots\dots\dots & \dots \\ a_{m1} & a_{m2} \dots \dots \dots \dots \dots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ .. \\ .. \\ .. \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ .. \\ .. \\ .. \\ b_m \end{pmatrix}$$

$$\boldsymbol{A} = \begin{pmatrix} a_{11} & a_{12} \dots \dots \dots \dots \dots & a_{1n} \\ a_{21} & a_{22} \dots \dots \dots \dots \dots & a_{2n} \\ \dots & \dots\dots\dots\dots\dots & \dots \\ \dots & \dots\dots\dots\dots\dots & \dots \\ \dots & \dots\dots\dots\dots\dots & \dots \\ a_{m1} & a_{m2} \dots \dots \dots \dots \dots & a_{mn} \end{pmatrix}, \quad \boldsymbol{X} = \begin{pmatrix} \boldsymbol{x_1} \\ \boldsymbol{x_2} \\ .. \\ .. \\ .. \\ \boldsymbol{x_n} \end{pmatrix} \quad and \quad \boldsymbol{B} = \begin{pmatrix} b_1 \\ b_2 \\ .. \\ .. \\ .. \\ b_m \end{pmatrix} \quad \text{where,}$$

$\boldsymbol{A}$ is called the ***matrix of coefficients***, $\boldsymbol{X}$ is the ***matrix of variables*** and $\boldsymbol{B}$ is the ***matrix of constants***.

Definition (1): *The system of linear equations,* $\boldsymbol{AX = B}$ *is called* ***nonhomogeneous system*** *If* $\boldsymbol{B \neq 0}$, *and is called* ***homogenous system*** $if\ \boldsymbol{AX = 0}$, $i.e.\ \boldsymbol{B = 0}$.

The linear system is represented as an ***augmented matrix :*** $(\boldsymbol{A}|\boldsymbol{B}) = \left(\begin{array}{ccc|c} a_{11} & a_{12} \dots \dots \dots \dots \dots & a_{1n} & b_1 \\ a_{21} & a_{22} \dots \dots \dots \dots \dots & a_{2n} & b_2 \\ \dots & \dots\dots\dots\dots\dots & \dots & .. \\ \dots & \dots\dots\dots\dots\dots & \dots & .. \\ \dots & \dots\dots\dots\dots\dots & \dots & .. \\ a_{m1} & a_{m2} \dots \dots \dots \dots \dots & a_{mn} & b_m \end{array}\right)$

This matrix is then modified using ***elementary row operations*** until it reaches ***reduced row echelon form***.

There are three types of elementary row operations:

1: **Swap the positions of two rows.**
2: Multiply a row by a nonzero scalar.
3: Add to one row a scalar multiple of another.

Because these operations are reversible, the augmented matrix produced always represents a linear system that is equivalent to the original.

Definition (2): A matrix $\boldsymbol{A}$ is said to be in ***echelon form*** if,

i) All the zero rows of $\boldsymbol{A}$ followed by non-zero row.
ii) The number of zero's before the first non-zero element in a non-zero row is less than the number of such zeros in the next row.
iii) The first non-zero element in a non-zero row is 1.

Definition (3): The ***rank of the matrix*** A is the number of non-zero rows in its echelon form.
There are several specific algorithms to row-reduce an augmented matrix, the simplest of which are ***Gaussian elimination*** ***and*** ***Gauss-Jordan elimination*** *and there is another one called* ***LU factorization.***

Definition (4): A solution of a linear system is an assignment of values to the variables $\boldsymbol{x_1}, \boldsymbol{x_2}, \dots \dots \dots, \boldsymbol{x_n}$ such that each of the equations is satisfied. The set of all possible solutions is called the *solution set.*
A linear system may behave in any one of three possible ways:

- The system has a single *unique solution*.
- The system has *infinitely many solutions*.
- The system has *no solution*.

Dentition (5): A linear system is called ***consistent*** *if it has at least one solution*, otherwise the system is said to be ***inconsistent.*** When the system is inconsistent, it is possible to derive a contradiction from the equations that may always be rewritten such as the statement 0 = 1.

Geometric interpretation

For a system involving two variables (*x* and *y*), each linear equation determines *a line* on the *xy*-plane. Because a solution to a linear system must satisfy all of the equations, the solution set is the intersection of these lines, and is hence either a line, a single point, or the empty set.

For three variables, each linear equation determines *a plane* in three-dimensional space, and the solution set is the intersection of these planes. Thus the solution set may be a plane, a line, a single point, or the empty set. For example, as three parallel planes do not have a common point, the solution set of their equations is empty; the solution set of the equations of three planes intersecting at a point is single point; if three planes pass through two points, their equations have at least two common solutions; in fact the solution set is infinite and consists in all the line passing through these points.

For *n* variables, each linear equation determines *a hyper plane in n-dimensional space*. The solution set is the intersection of these hyper planes, which may be a flat of any dimension.

Examples on Nonhomogeneous System, $AX = B$

Example (1): **Solve the following system of equations:**

$$x + 5y = 7$$
$$-2x - 7y = -5$$

i) Using the inverse matrix method, **ii) Using *Gauss-Jordan elimination method***

Solution

i) *Using the inverse matrix method.*

We can write the given system in the form: $AX = B \implies \begin{bmatrix} 1 & 5 \\ -2 & -7 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 7 \\ -5 \end{bmatrix}$

$\because \ |A| = 3 \neq 0, \ therefore \ A^{-1} = \frac{1}{|A|} adjA = \frac{1}{3}\begin{bmatrix} -7 & -5 \\ 2 & 1 \end{bmatrix}$

$\therefore \ A^{-1}(AX) = A^{-1}B \implies X = A^{-1}B \implies \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{3}\begin{bmatrix} -7 & -5 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 7 \\ -5 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} -24 \\ 9 \end{bmatrix} = \begin{bmatrix} -8 \\ 3 \end{bmatrix}$

$\therefore \ x = -8 \ , \ y = 3$

ii) *Using Gauss-Jordan elimination method*

$$\left(\begin{array}{cc|c} 1 & 5 & 7 \\ -2 & -7 & -5 \end{array}\right) \xrightarrow{2R_1 + R_2} \left(\begin{array}{cc|c} 1 & 5 & 7 \\ 0 & 3 & 9 \end{array}\right) \xrightarrow{\frac{1}{3}R_2} \left(\begin{array}{cc|c} 1 & 5 & 7 \\ 0 & 1 & 3 \end{array}\right) \xrightarrow{-5R_2 + R_1} \left(\begin{array}{cc|c} 1 & 0 & -8 \\ 0 & 1 & 3 \end{array}\right)$$

$\therefore$ The system is ***consistent*** and has only one solution, $x = -8 \ , \ y = 3$

Note: *The given system represents a pair of straight lines that they are intersecting at one point* $(-8, 3)$.

Example (2): **Solve the following system of equations:**

$$2x + 3y = 6$$
$$4x + 6y = 15$$

i) Using the inverse matrix method, **ii) Using *Gauss-Jordan elimination method***

Solution

i) *Using the inverse matrix method.*

$\because \ |A| = \begin{vmatrix} 2 & 3 \\ 4 & 6 \end{vmatrix} = 0$. *therefore*

A^{-1} **does not exict, so we can not solve** *using the inverse matrix method*

ii) *Using Gauss-Jordan elimination method*

$$\left(\begin{array}{cc|c} 2 & 3 & 6 \\ 4 & 6 & 15 \end{array}\right) \xrightarrow{-2R_1 + R_2} \left(\begin{array}{cc|c} 2 & 3 & 6 \\ 0 & 0 & 3 \end{array}\right) \implies 0 = 3, \quad \text{and this is contradiction.}$$

Therefore, the system is ***inconsistent*** and has no solution..

Note: *The given system represents a* ***pair of straight lines that they are parallel***.(**there is no solution**).

Example (3): **Solve the following system of equations:**

$$2x + 5y = 7$$
$$6x + 15y = 21$$

i) Using the inverse matrix method, **ii) Using *Gauss-Jordan elimination method***

Solution

i) *Using the inverse matrix method.*

$$\because\ |A| = \begin{vmatrix} 2 & 5 \\ 6 & 15 \end{vmatrix} = 0,\ therefore\ .$$

$$\boldsymbol{A^{-1}\ does\ not\ exict, so\ we\ can\ not\ solve}\ using\ the\ inverse\ matrix\ method.$$

ii) *Using Gauss-Jordan elimination method*

$$\left(\begin{array}{cc|c} 2 & 5 & 7 \\ 6 & 15 & 21 \end{array}\right) \xrightarrow{-3R_1 + R_2} \left(\begin{array}{cc|c} 2 & 5 & 7 \\ 0 & 0 & 0 \end{array}\right)$$

The system was reduced to one equation $2x + 5y = 7$, and it means that the two equations are identical.

$$\therefore\ The\ system\ \boldsymbol{is\ consistent}\ and\ has\ an\ infinite\ number\ of\ solutions.$$

Note: *The given system represents **a pair of identical straight lines** (**an infinite number of solutions**)*

Example (4): **Solve the following system of equations:**

$$x_1 - 2x_2 + 2x_3 = 5,$$
$$x_1 - x_2 = -1,$$
$$-x_1 + x_2 + x_3 = 5$$

i) Using the inverse matrix method, **ii) Using *Gauss-Jordan elimination method***

Solution

i) *Using the inverse matrix method.*

$$\because\ |A| = \begin{vmatrix} 1 & -2 & 2 \\ 1 & -1 & 0 \\ -1 & 1 & 1 \end{vmatrix} = 1\begin{vmatrix} -1 & 0 \\ 1 & 1 \end{vmatrix} + 2\begin{vmatrix} 1 & 0 \\ -1 & 1 \end{vmatrix} + 2\begin{vmatrix} 1 & -1 \\ -1 & 1 \end{vmatrix} = -1 + 2 + 0 = 1 \neq 0\ \text{,therefore}$$

$$A^{-1} = \frac{1}{|A|} adjA = \begin{bmatrix} -1 & -1 & 0 \\ 4 & 3 & 1 \\ 2 & 2 & 1 \end{bmatrix}^T = \begin{bmatrix} -1 & 4 & 2 \\ -1 & 3 & 2 \\ 0 & -3 & 1 \end{bmatrix}$$

We can write the system in the form: $AX = B \implies \begin{bmatrix} 1 & -2 & 2 \\ 1 & -1 & 0 \\ -1 & 1 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5 \\ -1 \\ 5 \end{bmatrix}$

$$\therefore\ X = A^{-1}B \implies \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 & 4 & 2 \\ -1 & 3 & 2 \\ 0 & 1 & 1 \end{bmatrix}\begin{bmatrix} 5 \\ -1 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}\ therefore,$$

$$\therefore\ \quad x_1 = 1 \quad , \quad x_2 = 2 \quad , \quad x_3 = 4$$

Note: *The given system represents three planes and they are intersecting at one point* $(1, 2, 4,)$.

ii) *Using Gauss-Jordan elimination method*

$$\left(\begin{array}{ccc|c} 1 & -2 & 2 & 5 \\ 1 & -1 & 0 & -1 \\ -1 & 1 & 1 & 5 \end{array}\right) \xrightarrow{-1R_1+R_2} \left(\begin{array}{ccc|c} 1 & -2 & 2 & 5 \\ 0 & 1 & -2 & -6 \\ -1 & 1 & 1 & 5 \end{array}\right) \xrightarrow{R_1+R_3} \left(\begin{array}{ccc|c} 1 & -2 & 2 & 5 \\ 0 & 1 & -2 & -6 \\ 0 & -1 & 3 & 10 \end{array}\right)$$

$$\xrightarrow{R_2+R_3} \left(\begin{array}{ccc|c} 1 & -2 & 2 & 5 \\ 0 & 1 & -2 & -6 \\ 0 & 0 & 1 & 4 \end{array}\right) \xrightarrow{2R_3+R_2} \left(\begin{array}{ccc|c} 1 & -2 & 2 & 5 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 4 \end{array}\right) \xrightarrow{2R_2+R_1} \left(\begin{array}{ccc|c} 1 & 0 & 2 & 9 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 4 \end{array}\right)$$

$$\xrightarrow{-2R_3+R_1} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 4 \end{array}\right) \quad \Longrightarrow \quad \textit{The system } \textbf{is consistent} \textit{ and has only } \textbf{one solution},$$

$\textit{Therefore},\quad x_1 = 1 \quad , \quad x_2 = 2 \quad , \quad x_3 = 4$

Example (5): **Find all solutions to the following systems of linear equations;**

$$\begin{aligned} x_1 - x_2 + 2x_3 &= 1 \\ 2x_1 + x_2 + x_3 &= 8 \\ x_1 + x_2 \quad\quad &= 5, \end{aligned}$$

i) Using the inverse matrix method, **ii) Using *Gauss-Jordan elimination method***

Solution

i) *Using the inverse matrix method*

$\because \ |A| = \begin{vmatrix} 1 & -1 & 2 \\ 2 & 1 & 1 \\ 1 & 1 & 0 \end{vmatrix} = 1\begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} + 1\begin{vmatrix} 2 & 1 \\ 1 & 0 \end{vmatrix} + 2\begin{vmatrix} 2 & 1 \\ 1 & 1 \end{vmatrix} = -1 - 1 + 2 = 0$ *,therefore*

A^{-1} ***does not exict, so we can not solve*** *using the inverse matrix method.*

ii) *Using Gauss-Jordan elimination method*

$$\left(\begin{array}{ccc|c} 1 & -1 & 2 & 1 \\ 2 & 1 & 1 & 8 \\ 1 & 1 & 0 & 5 \end{array}\right) \xrightarrow{-2R_1+R_2} \left(\begin{array}{ccc|c} 1 & -1 & 2 & 1 \\ 0 & 3 & -3 & 6 \\ 1 & 1 & 0 & 5 \end{array}\right) \xrightarrow{R_1+R_3} \left(\begin{array}{ccc|c} 1 & -1 & 2 & 1 \\ 0 & 3 & -3 & 6 \\ 0 & 2 & -2 & 4 \end{array}\right) \xrightarrow{\frac{1}{3}R_2}$$

$$\left(\begin{array}{ccc|c} 1 & -1 & 2 & 1 \\ 0 & 1 & -1 & 2 \\ 0 & 2 & -2 & 4 \end{array}\right) \xrightarrow{\frac{1}{2}R_3} \left(\begin{array}{ccc|c} 1 & -1 & 2 & 1 \\ 0 & 1 & -1 & 2 \\ 0 & 1 & -1 & 2 \end{array}\right) \xrightarrow{-R_2+R_3} \left(\begin{array}{ccc|c} 1 & -1 & 2 & 1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right) \xrightarrow{-R_2+R_1}$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 3 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right),$$

Since the last row contains only zeros, this means that the system is reduced to two equations containing three unknowns therefore, the system ***is consistent*** and **has an infinite number of solutions.**

To get these solutions, $\textit{put} \quad x_3 = r \quad (r \textit{ is real number}), \textit{then}$

From the 2-nd row we get, $\quad x_2 - x_3 = 2 \quad \Longrightarrow \quad x_2 = x_3 + 2 \quad \Longrightarrow \quad x_2 = r + 2$ ***and,***

From the 1-st row we get, $\quad x_1 + x_3 = 3 \quad \Longrightarrow \quad x_1 = 3 - x_3 \quad \Longrightarrow \quad x_1 = 3 - r$

Therefore, the solution is: $(x\ ,\ y\ ,\ z) = (3 - r\ ,\ r + 2\ ,\ r)$

Note:

1) Since the last row contains only zeros, this means also that the matrix **A** is singular matrix ($|A| = \mathbf{0}$) and the system represents three equations that are intersecting in straight line,so there are an infinite number of solutions.(an infinite number of points that are common points on the three planes.

2) ***The rank of matrix A*** is two,where the number of nonzero rows is two after applying a set of elementary row operations on **A.**

Example (6): **Solve the following system of linear equations;**

$$\begin{aligned} x_1 + \; x_2 + 3x_3 &= 3 \\ -x_1 + \; x_2 + \; x_3 &= -1 \\ 2x_1 + 3x_2 + 8\,x_3 &= 4, \end{aligned}$$

i) Using the inverse matrix method, **ii) Using *Gauss-Jordan elimination method***

Solution

i) *Using the inverse matrix method*

$$\because \; |A| = \begin{vmatrix} 1 & 1 & 3 \\ -1 & 1 & 1 \\ 2 & 3 & 8 \end{vmatrix} = 1\begin{vmatrix} 1 & 1 \\ 3 & 8 \end{vmatrix} - 1\begin{vmatrix} -1 & 1 \\ 2 & 8 \end{vmatrix} + 3\begin{vmatrix} -1 & 1 \\ 2 & 3 \end{vmatrix} = 5 + 10 - 15 = 0 \text{ ,therefore}$$

A^{-1} **does not exict, so we can not solve** *using the inverse matrix method.*

ii) *Using Gauss-Jordan elimination method*

$$\left(\begin{array}{ccc|c} 1 & 1 & 3 & 3 \\ -1 & 1 & 1 & -1 \\ 2 & 3 & 8 & 4 \end{array}\right) \xrightarrow{R_1 + R_2} \left(\begin{array}{ccc|c} 1 & 1 & 3 & 3 \\ 0 & 2 & 4 & 2 \\ 2 & 3 & 8 & 4 \end{array}\right) \xrightarrow{-2R_1 + R_3} \left(\begin{array}{ccc|c} 1 & 1 & 3 & 3 \\ 0 & 2 & 4 & 2 \\ 0 & 1 & 2 & -2 \end{array}\right)$$

$$\xrightarrow{\frac{1}{2}R_2} \left(\begin{array}{ccc|c} 1 & 1 & 3 & 3 \\ 0 & 1 & 2 & 1 \\ 0 & 1 & 2 & -2 \end{array}\right) \xrightarrow{-R_2 + R_3} \left(\begin{array}{ccc|c} 1 & 1 & 3 & 3 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & -3 \end{array}\right),$$

The last row shows that, $\mathbf{0 = 3,}$ and ***this is contradiction***, therefore, the system is ***inconsistent*** and **has no solution**. (there is no common point satisfies the three equations).

Example (7): **Solve the following system of equations.**

$$\begin{aligned} x + 2y + 8z - 7w &= -2 \\ 3x + 2y + 12z - 5w &= 6 \\ -x + \; y + \; z - 5w &= -10 \end{aligned}$$

Solution

The augmented matrix of the system of equations is:

$$\left(\begin{array}{cccc|c} 1 & 2 & 8 & -7 & -2 \\ 3 & 2 & 12 & -5 & 6 \\ -1 & 1 & 1 & -5 & -10 \end{array}\right) \xrightarrow{-3R_1 + R_2} \left(\begin{array}{cccc|c} 1 & 2 & 8 & -7 & -2 \\ 0 & -4 & -12 & 16 & 12 \\ -1 & 1 & 1 & -5 & -10 \end{array}\right) \xrightarrow{R_1 + R_3}$$

$$\left(\begin{array}{cccc|c} 1 & 2 & 8 & -7 & -2 \\ 0 & -4 & -12 & 16 & 12 \\ 0 & 3 & 9 & -12 & -12 \end{array}\right) \xrightarrow{-\frac{1}{4}R_2} \left(\begin{array}{cccc|c} 1 & 2 & 8 & -7 & -2 \\ 0 & 1 & 3 & -4 & -3 \\ 0 & 3 & 9 & -12 & 12 \end{array}\right) \xrightarrow{-\frac{1}{3}R_3}$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & -2 \\ 0 & 1 & 3 & -4 & 12 \\ 0 & -1 & -3 & 4 & 12 \end{array}\right) \xrightarrow{R_2+R_3} \left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & -2 \\ 0 & 1 & 3 & -4 & 12 \\ 0 & 0 & 0 & 0 & 24 \end{array}\right)$$

The last row shows that, $0 = 24$, *and* **this is contradiction**, *so the system* **has no solution**.

Note: **The rank of matrix A** ***is two***, where the number of nonzero rows is two after applying a set of elementary row operations on **A.**

LU decomposition of a matrix

The ***LU decomposition,*** also called ***LU factorization*** (where ' ***LU*** ' stands for ***'lower upper'***) is another approach designed to exploit triangular systems. The LU decomposition was introduced by the famous mathematician Alan Turing in 1948.

The **LU decomposition of a matrix** A is the product of a lower triangular matrix and an upper triangular matrix that is equal to A. We suppose that we can write $A = LU$, where L is a lower triangular matrix and U is an upper triangular matrix.

Our aim is to find L and U and once we have done so we have found an LU decomposition of A.

Example (1): Find an ***LU*** decomposition of $A = \begin{bmatrix} 3 & 1 \\ -6 & -4 \end{bmatrix}$

Solution

$$\textit{Let},\ A = \begin{bmatrix} 3 & 1 \\ -6 & -4 \end{bmatrix} = LU = \begin{bmatrix} 1 & 0 \\ L_{21} & 1 \end{bmatrix}\begin{bmatrix} U_{11} & U_{12} \\ 0 & U_{22} \end{bmatrix}$$

$$\begin{bmatrix} U_{11} & U_{12} \\ L_{21}U_{11} & L_{21}U_{12} + U_{22} \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ -6 & -4 \end{bmatrix}$$

From the 1st. row we get: $U_{11} = 3$ **and** $U_{12} = 1$

From the 2nd. row we get: $L_{21}U_{11} = -6 \Rightarrow L_{21} = -2$ *and,*

$L_{21}U_{12} + U_{22} = -4 \Rightarrow U_{22} = -2$

Therefore,

$$A = \begin{bmatrix} 3 & 1 \\ -6 & -4 \end{bmatrix} = LU = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}\begin{bmatrix} 3 & 1 \\ 0 & -2 \end{bmatrix}$$

Example (2): Find an LU decomposition of $A = \begin{bmatrix} 1 & 2 & 4 \\ 3 & 8 & 14 \\ 2 & 6 & 13 \end{bmatrix}$.

Solution

Let, $A = \begin{bmatrix} 1 & 2 & 4 \\ 3 & 8 & 14 \\ 2 & 6 & 13 \end{bmatrix} = LU = \begin{bmatrix} 1 & 0 & 0 \\ L_{21} & 1 & 0 \\ L_{31} & L_{32} & 1 \end{bmatrix} \begin{bmatrix} U_{11} & U_{12} & U_{13} \\ 0 & U_{22} & U_{23} \\ 0 & 0 & U_{33} \end{bmatrix}$

Multiplying out LU and setting the answer equal to A gives;

$$\begin{bmatrix} U_{11} & U_{12} & U_{13} \\ L_{21}U_{11} & L_{21}U_{12} + U_{22} & L_{21}U_{13} + U_{23} \\ L_{31}U_{11} & L_{31}U_{12} + L_{32}U_{22} & L_{31}U_{13} + L_{32}U_{23} + U_{33} \end{bmatrix} = \begin{bmatrix} 1 & 2 & 4 \\ 3 & 8 & 14 \\ 2 & 6 & 13 \end{bmatrix}$$

From the 1st. row we get: $U_{11} = 1$, $U_{12} = 2$, $U_{13} = 4$

From the 2nd. row we get: $L_{21}U_{11} = 3 \quad \Rightarrow \quad L_{21} = 3$ **and,**

$L_{21}U_{12} + U_{22} = 8 \quad \Rightarrow \quad U_{22} = 2$

$L_{21}U_{13} + U_{23} = 14 \quad \Rightarrow \quad U_{23} = 2$

From the 2nd. row we get: $L_{31}U_{11} = 2 \quad \Rightarrow \quad L_{31} = 2$

$L_{31}U_{12} + L_{32}U_{22} = 6 \quad \Rightarrow \quad L_{32} = 1$

$L_{31}U_{13} + L_{32}U_{23} + U_{33} = 13 \quad \Rightarrow \quad U_{33} = 3$

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 3 & 8 & 14 \\ 2 & 6 & 13 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 2 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 4 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}$$

Using an *LU* decomposition to solve systems of equations

Once a matrix A has been decomposed into lower and upper triangular parts it is possible to obtain the solution to $AX = B$ in a direct way.

The procedure can be summarized as follows;

- Given A, find L and U so that $A = LU$. Hence $LUX = B$.
- Let $Y = UX$ so that $LY = B$. Solve this triangular system for Y
- Finally solve the triangular system $UX = Y$ for X.

Example (3): Solve the following linear system of equations using the LU factorization method

$$x_1 + 2\,x_2 + \ 4x_3 = 3$$

$$3x_1 + 8x_2 + 14x_3 = 13$$

$$2x_1 + 6\,x_2 + 13x_3 = 4$$

Solution

- The first step is to calculate the ***LU*** decomposition of the coefficient matrix A
- In this case that job has already been done in the previous example (**Example 2**). We found that

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 2 & 1 & 1 \end{bmatrix} \qquad and \qquad U = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}$$

- The next step is to solve $LY = B$ for the vector $Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$. That is we consider

$$LY = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 2 & 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 13 \\ 4 \end{bmatrix} = B \qquad \Longrightarrow \qquad \begin{aligned} y_1 + 0 + 0 &= 3 \\ 3y_1 + y_2 + 0 &= 13 \\ 2y_1 + y_2 + y_3 &= 4 \end{aligned}$$

From the top equation we see that: $y_1 = 3$.

The middle equation states that: $3y_1 + y_2 = 13 \quad \Longrightarrow \quad y_2 = 4$.

Finally the bottom line says that $2y_1 + y_2 + y_3 = 4 \quad \Longrightarrow \quad y_3 = -6$

- $$UX = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ -6 \end{bmatrix} = Y \qquad \Longrightarrow \qquad \begin{aligned} x_1 + 2x_2 + 4x_3 &= 3 \\ 0 + 2x_2 + 2x_3 &= 4 \\ 0 + 0 + 3x_3 &= -6 \end{aligned}$$

Starting with the bottom equation we see that: $3x_3 = -6 \quad \Longrightarrow \quad x_3 = -2$

The middle equation implies that: $2x_2 + 2x_3 = 4 \quad \Longrightarrow \quad x_2 = 4$

The top equation states that: $x_1 + 2x_2 + 4x_3 = 3 \quad \Longrightarrow \quad x_1 = 3$

Therefore we have found that the solution to the system of simultaneous equations $AX = B$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ -2 \end{bmatrix}$$

Do matrices always have an LU decomposition? No

Sometimes it is impossible to write a matrix in the form "lower triangular" × "upper triangular". **Why not?**

An invertible matrix ***A*** has an ***LU*** **decomposition** provided that all its **leading submatrices** have non-zero determinants. The k-th leading submatrix of A is denoted $\mathbf{A}_k$ and is the $k \times k$ matrix found by looking only at the top k rows and left most k columns.

For example, if

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 3 & 8 & 14 \\ 2 & 6 & 13 \end{bmatrix}$$

then the leading submatrices are

$$A_1 = [1] \quad , \quad A_2 = \begin{bmatrix} 1 & 2 \\ 3 & 8 \end{bmatrix} \quad , \quad A_3 = \begin{bmatrix} 1 & 2 & 4 \\ 3 & 8 & 14 \\ 2 & 6 & 13 \end{bmatrix}$$

The fact that this matrix ***A*** has an ***LU*** **decomposition** can be guaranteed in advance because none of these determinants is zero:

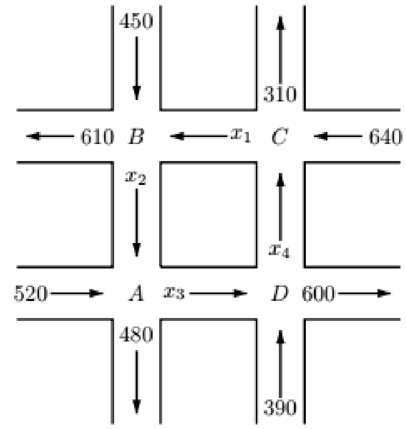
$$|A_1| = |1| = 1 \neq 0 \quad , \quad |A_2| = \begin{vmatrix} 1 & 2 \\ 3 & 8 \end{vmatrix} = 2 \neq 0 \quad , \quad |A_3| = \begin{vmatrix} 1 & 2 & 4 \\ 3 & 8 & 14 \\ 2 & 6 & 13 \end{vmatrix} = 6 \neq 0$$

Example (4): Show that the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 1 & 3 & 4 \end{bmatrix}$ does not have an *LU* decomposition.

Solution

The second leading submatrix A_2 has determinant equal to $|A_2| = \begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix} = 0,$ which means that an *LU* decomposition is not possible in this case.

Application:



The flow of traffic (in vehicles per hour) through a network of streets is shown in the opposite figure.

The linear system that represents this network is given as follows:

$$x_2 + 520 = x_3 + 480$$
$$x_1 + 450 = x_2 + 610$$
$$x_4 + 640 = x_1 + 310$$
$$x_3 + 390 = x_4 + 600$$

Solve the given system and find the traffic flow when, $x_4 = 100$.

Solution

The given linear system may be written in the following form;

$$\begin{aligned} x_1 - x_2 + 0 + 0 &= 160 \\ -x_1 + 0 + 0 + x_4 &= -330 \\ 0 + x_2 - x_3 + 0 &= -40 \\ 0 + 0 + x_3 - x_4 &= 210 \end{aligned}$$

$$\left(\begin{array}{cccc|c} 1 & -1 & 0 & 0 & 160 \\ -1 & 0 & 0 & 1 & -330 \\ 0 & 1 & -1 & 0 & -40 \\ 0 & 0 & 1 & -1 & 210 \end{array}\right) \xrightarrow{R_1 + R_2} \left(\begin{array}{cccc|c} 1 & -1 & 0 & 0 & 160 \\ 0 & -1 & 0 & 1 & -170 \\ 0 & 1 & -1 & 0 & -40 \\ 0 & 0 & 1 & -1 & 210 \end{array}\right) \xrightarrow{R_2 + R_3}$$

$$\left(\begin{array}{cccc|c} 1 & -1 & 0 & 0 & 160 \\ 0 & -1 & 0 & 1 & -170 \\ 0 & 0 & -1 & 1 & -210 \\ 0 & 0 & 1 & -1 & 210 \end{array}\right) \xrightarrow{R_3 + R_4} \left(\begin{array}{cccc|c} 1 & -1 & 0 & 0 & 160 \\ 0 & -1 & 0 & 1 & -170 \\ 0 & 0 & -1 & 1 & -210 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

Since, the last row is zero's therefore, the system has an infinite number of solutions;

Let $\boldsymbol{x_4 = r}$, then $\boldsymbol{x_3 = r + 210}$, $\boldsymbol{x_2 = r + 170}$, $\boldsymbol{x_1 = r + 310}$

When $\boldsymbol{x_4 = 100}$, $\boldsymbol{x_3 = 310}$, $\boldsymbol{x_2 = 270}$, $\boldsymbol{x_1 = 410}$